Project Summary Report

Project summary report

This report combines the current Phase 2, Phase 3, and Phase 4 results.

Method note: the current completed workflow uses all 2,880 rows and a LightGBM ablation; it should be described as an updated implementation of the original proposal, not an exact execution of it.

Phase 2 headline results:

- S1_VH: best model is Embeddings only with R2=0.942, RMSE=1.182, MAE=0.858

- S1_VV: best model is Embeddings only with R2=0.902, RMSE=1.255, MAE=0.825

- S1_VV_div_VH: best model is Embeddings + region + Dynamic World with R2=0.698, RMSE=1.155, MAE=0.798

Phase 3 headline results for S1_VV:

Hardest land-use classes:

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built: MAE=1.227, bias=-0.333

water: MAE=1.011, bias=0.020

shrub_and_scrub: MAE=0.902, bias=-0.054
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Exploratory spatial signal only: california_coast has Moran's I=0.092 with p=0.026; this should not be presented as a strong confirmed clustering result.

Phase 4 headline results:

- Overall cross-modal neighbor overlap at k=10 is 0.032 versus a random-match baseline of about 0.003.

- Best land-use overlap at k=10 occurs for water (0.098).

- Overall embedding-to-SAR distance correlation is positive but modest: Pearson r=0.195, Spearman rho=0.151.

- Within-region correlation is much stronger, peaking in california_coast at Pearson

Best Phase 2 Models by Target

target	feature_set	r2	rmse	mae	pearson_r
S1_VH	Embeddings only	0.942	1.182	0.858	0.971
S1_VV	Embeddings only	0.902	1.255	0.825	0.951
S1_VV_div_VH	Embeddings + region + Dynamic World	0.698	1.155	0.798	0.836

These are the best held-out full-dataset models selected for the project summary.

Top S1_VV Failure Modes

dw_label_name	mae	rmse	bias	p90_abs_error	n
built	1.227	2.070	-0.333	2.176	96
water	1.011	1.500	0.020	1.957	96
shrub_and_scrub	0.902	1.366	-0.054	1.742	96
bare	0.837	1.202	0.289	1.697	96
crops	0.772	1.006	0.056	1.594	96

The cumulative report focuses on the highest-error S1_VV land-use classes.

S1_VV Spatial Residual Structure

region	morans_i	p_value	significant_at_0_05	n
amazon_forest	-0.026	0.506	False	216
california_coast	0.092	0.026	True	216
iowa_ag	0.012	0.742	False	216
sf_bay_urban	0.032	0.384	False	216

Significant spatial clustering indicates geographically structured failure modes.

Phase 4 Cross-modal Similarity

k	n_queries	mean_overlap	median_overlap	p10_overlap	p90_overlap
5	2880	0.020	0.000	0.000	0.000
10	2880	0.032	0.000	0.000	0.100
20	2880	0.053	0.000	0.000	0.150

Cross-modal similarity is summarized by k-NN overlap between embedding space and SAR space.

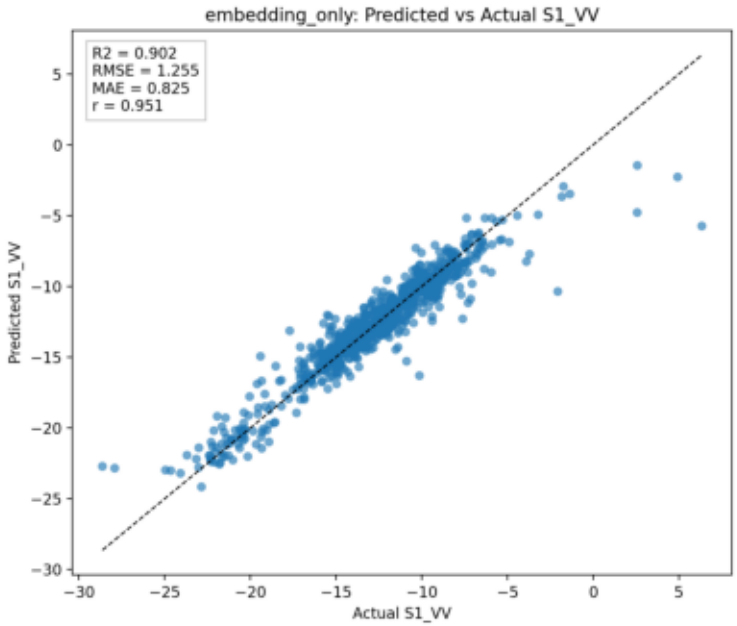
Top Phase 4 Land-use Alignment at k=10

dw_label_name	mean_overlap	median_overlap	p10_overlap	p90_overlap	n_queries
water	0.098	0.100	0.000	0.300	320
snow_and_ice	0.043	0.000	0.000	0.100	320
flooded_vegetation	0.036	0.000	0.000	0.100	320
trees	0.024	0.000	0.000	0.100	320
bare	0.022	0.000	0.000	0.100	320

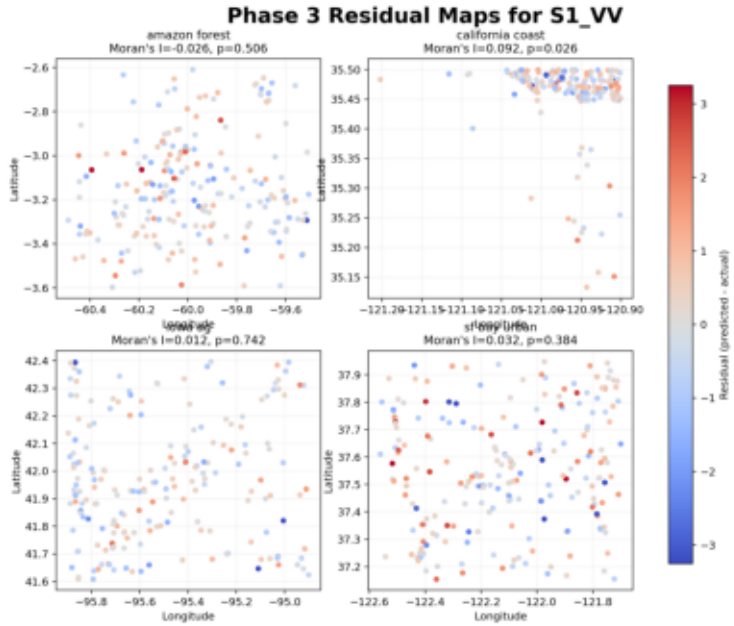
This table highlights which land-use classes show the strongest cross-modal neighbor agreement.

Project Visual Summary

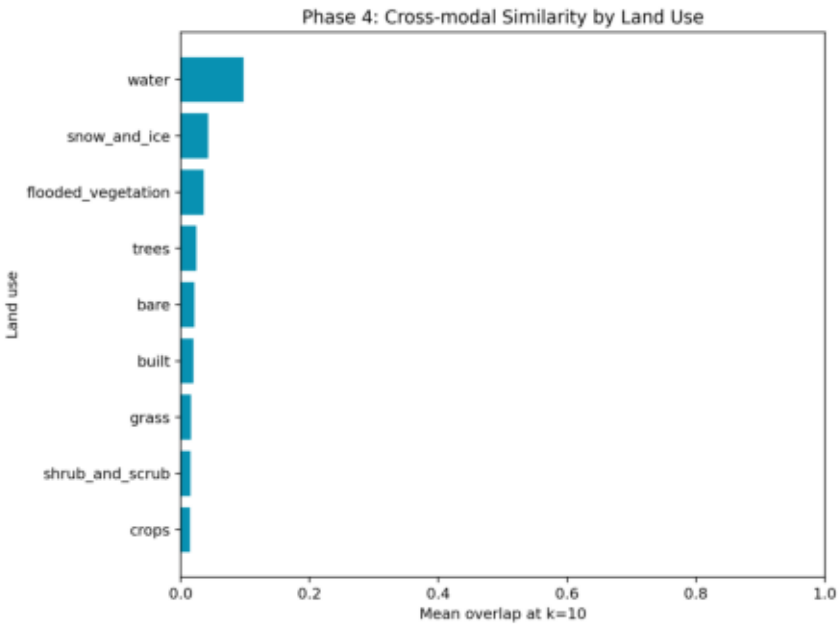
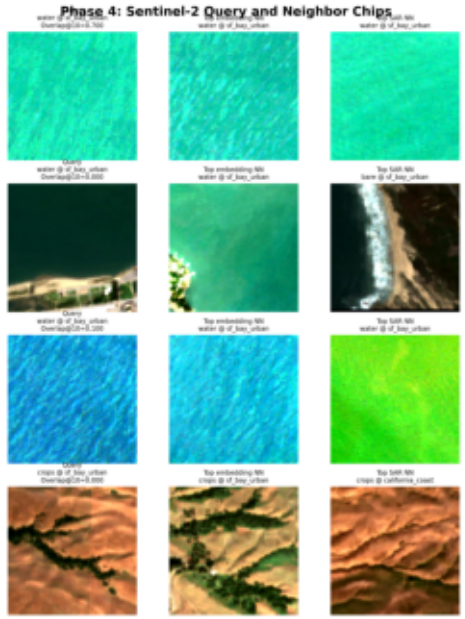
predicted vs actual embedding only S1 VV



phase3 residual maps S1 VV



phase4 knn overlap by land use k10



phase4 sentinel2 neighbor chips